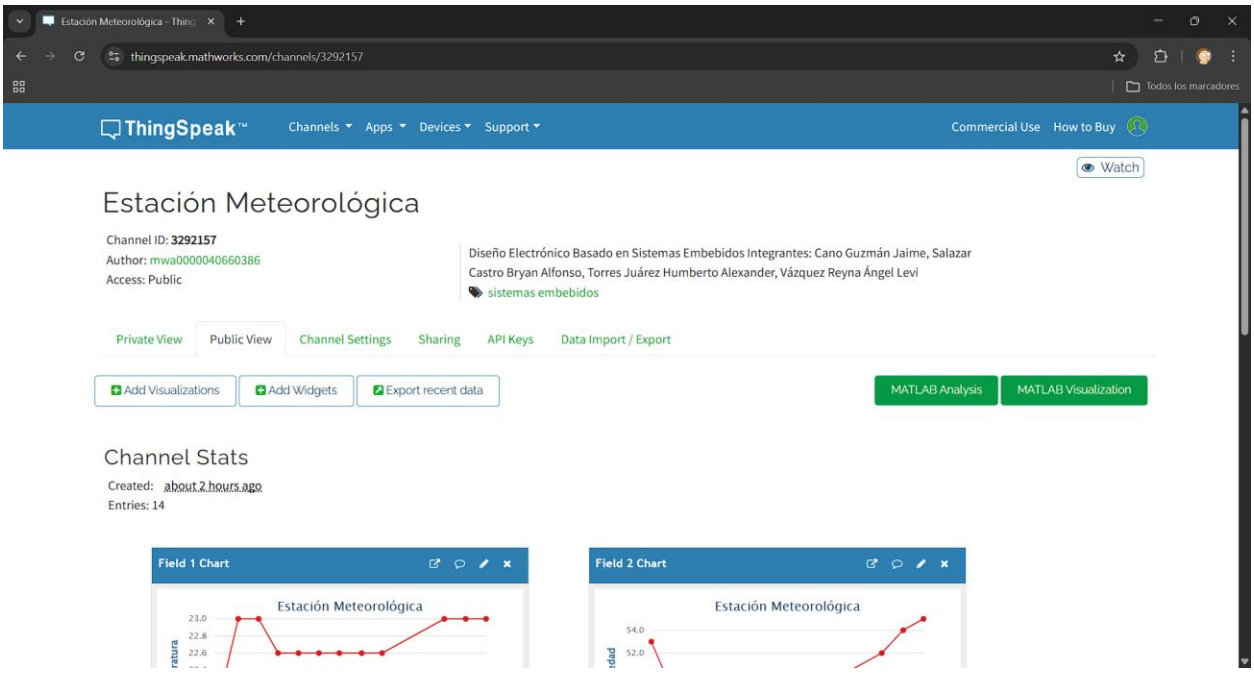

Estación Meteorológica - Thing
thingspeak.mathworks.com/channels/3292157
Todos los marcadores
ThingSpeak™
Channels
Apps
Devices
Support
Commercial Use
How to Buy
Watch
Estación Meteorológica
Channel ID: 3292157
Author: mwa0000040660386
Access: Public
Diseño Electrónico Basado en Sistemas Embebidos Integrantes: Cano Guzmán Jaime, Salazar Castro Bryan Alfonso, Torres Juárez Humberto Alexander, Vázquez Reyna Ángel Levi
sistemas embebidos
Private View
Public View
Channel Settings
Sharing
API Keys
Data Import / Export
Add Visualizations
Add Widgets
Export recent data
MATLAB Analysis
MATLAB Visualization
Channel Stats
Created: about 2 hours ago
Entries: 14
Field 1 Chart
Estación Meteorológica
23.0
22.8
22.6
Field 2 Chart
Estación Meteorológica
54.0
52.0

API Keys - ThingSpeak IoT

thingspeak.mathworks.com/channels/3292157/api_keys

ThingSpeak™ Channels Apps Devices Support Commercial Use How to Buy

Private View | Public View | Channel Settings | Sharing | API Keys | Data Import / Export

Write API Key

Key: 99NHFI8FYU3IMA5O

Generate New Write API Key

Read API Keys

Key: NHI8N9LCPOAF65ES

Note:

Save Note | Delete API Key

Add New Read API Key

Help

API keys enable you to write data to a channel or read data from a private channel. API keys are auto-generated when you create a new channel.

API Keys Settings

- **Write API Key**: Use this key to write data to a channel. If you feel your key has been compromised, click **Generate New Write API Key.**
- **Read API Keys**: Use this key to allow other people to view your private channel feeds and charts. Click **Generate New Read API Key** to generate an additional read key for the channel.
- **Note**: Use this field to enter information about channel read keys. For example, add notes to keep track of users with access to your channel.

API Requests

Write a Channel Feed

```
GET https://api.thingspeak.com/update?api_key=99NHFI8FYU3IMA5O&field1=0
```

Read a Channel Feed

```
GET https://api.thingspeak.com/channels/3292157/feeds.json?api_key=NHI8N9LCPOAF65E
```

Read a Channel Field

```
GET https://api.thingspeak.com/channels/3292157/fields/1.json?api_key=NHI8N9LCPOAF
```

File Edit Selection View Go Run Terminal Help — EstacionMeteorologica

EXPLORER — ESTACIONMETEOROLOGICA

- .pio
- .vscode
- include
- lib
- src
- test
- .gitignore
- platformio.ini

platformio.ini

```
; PlatformIO Project Configuration File
;
;   Build options: build flags, source filter
;   Upload options: custom upload port, speed and extra flags
;   Library options: dependencies, extra library storages
;   Advanced options: extra scripting
;
; Please visit documentation for the other options and examples
; https://docs.platformio.org/page/projectconf.html

[env:esp32dev]
platform = espressif32
board = esp32dev
framework = arduino
lib_deps =
    adafruit/Adafruit Unified Sensor @ ^1.1.15
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe run
========================================= [SUCCESS] Took 15.34 seconds =========================================
Terminal will be reused by tasks, press any key to close it.
```

CHAT

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

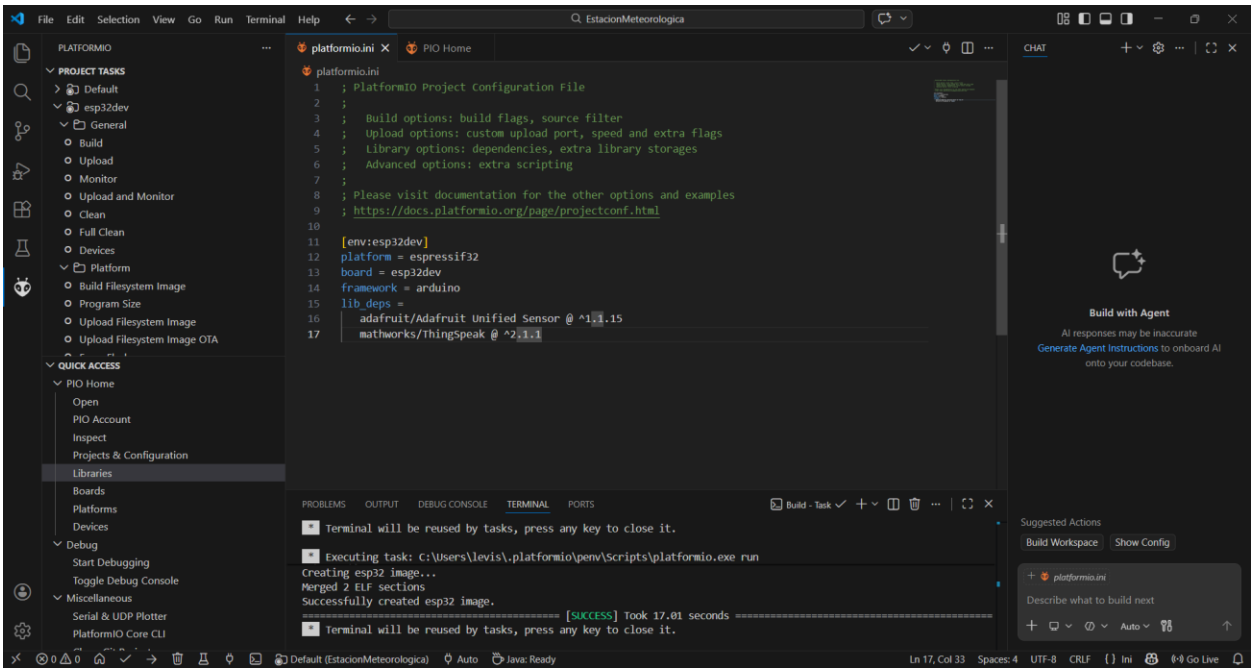
Suggested Actions: Build Workspace | Show Config

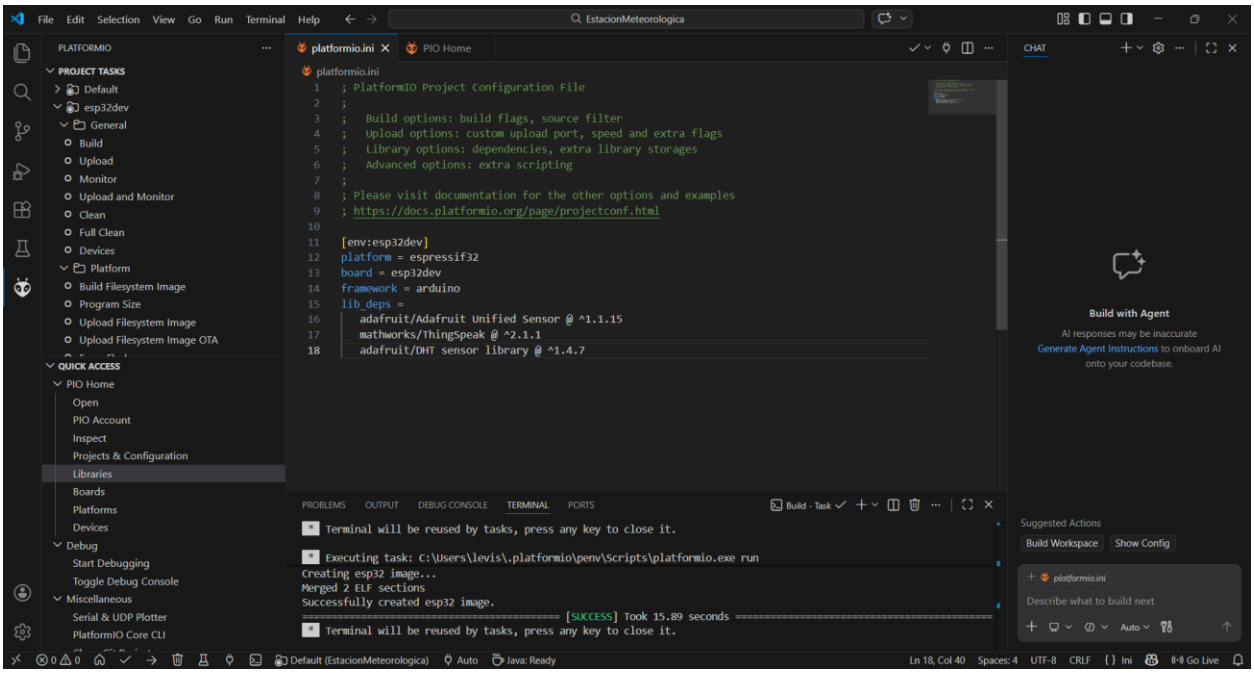
platformio.ini

Describe what to build next

OUTLINE | TIMELINE | JAVA PROJECTS

Default (EstacionMeteorologica) | Auto | Java: Ready | Ln 16, Col 5 | Spaces: 4 | UTF-8 | CRLF | Ini | Go Live


File Edit Selection View Go Run Terminal Help
EstacionMeteorologica
PLATFORMIO
PROJECT TASKS
Default
esp32dev
General
Build
Upload
Monitor
Upload and Monitor
Clean
Full Clean
Devices
Platform
Build Filesystem Image
Program Size
Upload Filesystem Image
Upload Filesystem Image OTA
QUICK ACCESS
PIO Home
Open
PIO Account
Inspect
Projects & Configuration
Libraries
Boards
Platforms
Devices
Debug
Start Debugging
Toggle Debug Console
Miscellaneous
Serial & UDP Plotter
PlatformIO Core CLI
platformio.ini
PIO Home
platformio.ini
; PlatformIO Project Configuration File
;
; Build options: build flags, source filter
; Upload options: custom upload port, speed and extra flags
; Library options: dependencies, extra library storages
; Advanced options: extra scripting
;
; Please visit documentation for the other options and examples
; https://docs.platformio.org/page/projectconf.html
[env:esp32dev]
platform = espressif32
board = esp32dev
framework = arduino
lib_deps =
adafruit/Adafruit Unified Sensor @ ^1.1.15
mathworks/ThingSpeak @ ^2.1.1
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Build - Task
Terminal will be reused by tasks, press any key to close it.
Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe run
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
========== [SUCCESS] Took 17.01 seconds ==========
Terminal will be reused by tasks, press any key to close it.
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Suggested Actions
Build Workspace
Show Config
platformio.ini
Describe what to build next
Auto
Default (EstacionMeteorologica)
Auto
Java: Ready
Ln 17, Col 33
Spaces: 4
UTF-8
CRLF
Ini
Go Live


File Edit Selection View Go Run Terminal Help
EstacionMeteorologica
PLATFORMIO
PROJECT TASKS
Default
esp32dev
General
Build
Upload
Monitor
Upload and Monitor
Clean
Full Clean
Devices
Platform
Build Filesystem Image
Program Size
Upload Filesystem Image
Upload Filesystem Image OTA
QUICK ACCESS
PIO Home
Open
PIO Account
Inspect
Projects & Configuration
Libraries
Boards
Platforms
Devices
Debug
Start Debugging
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Miscellaneous
Serial & UDP Plotter
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platformio.ini
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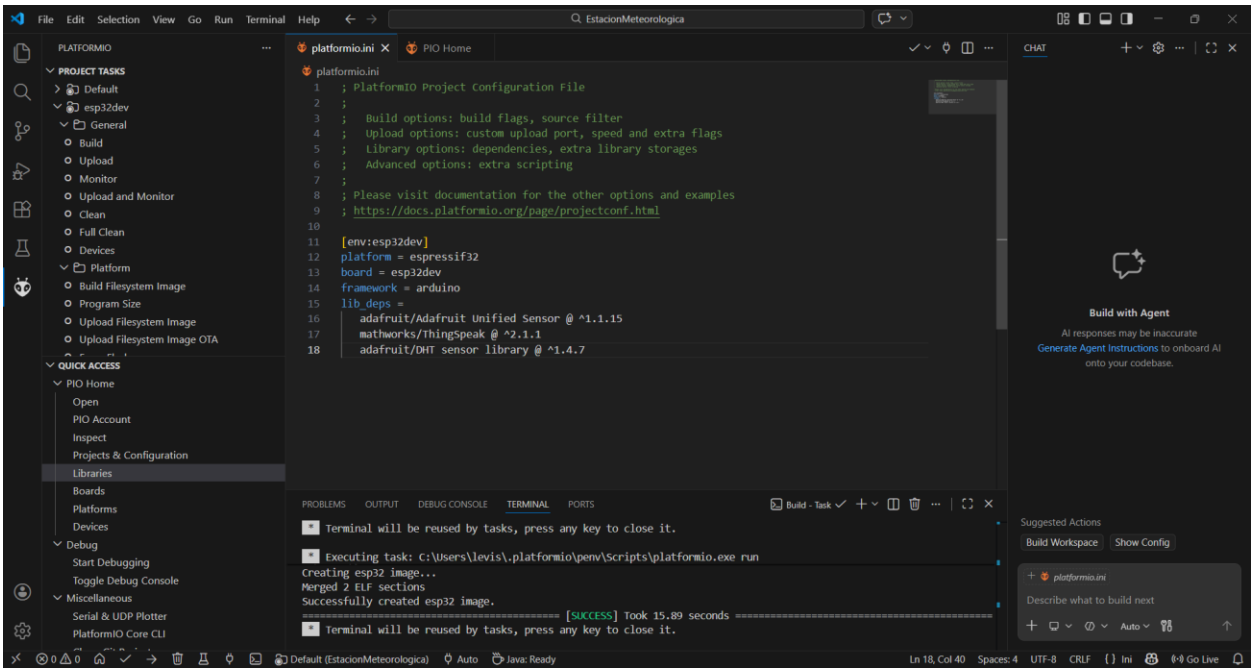
; https://docs.platformio.org/page/projectconf.html
[env:esp32dev]
platform = espressif32
board = esp32dev
framework = arduino
lib_deps =
adafruit/Adafruit Unified Sensor @ ^1.1.15
mathworks/ThingSpeak @ ^2.1.1
adafruit/DHT sensor library @ ^1.4.7
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Build - Task
Terminal will be reused by tasks, press any key to close it.
Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe run
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
========== [SUCCESS] Took 15.89 seconds ==========
Terminal will be reused by tasks, press any key to close it.
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Suggested Actions
Build Workspace
Show Config
platformio.ini
Describe what to build next
Auto
Default (EstacionMeteorologica)
Auto
Java: Ready
Ln 18, Col 40
Spaces: 4
UTF-8
CRLF
Ini
Go Live


platformio.ini
; PlatformIO Project Configuration File
;
; Build options: build flags, source filter
; Upload options: custom upload port, speed and extra flags
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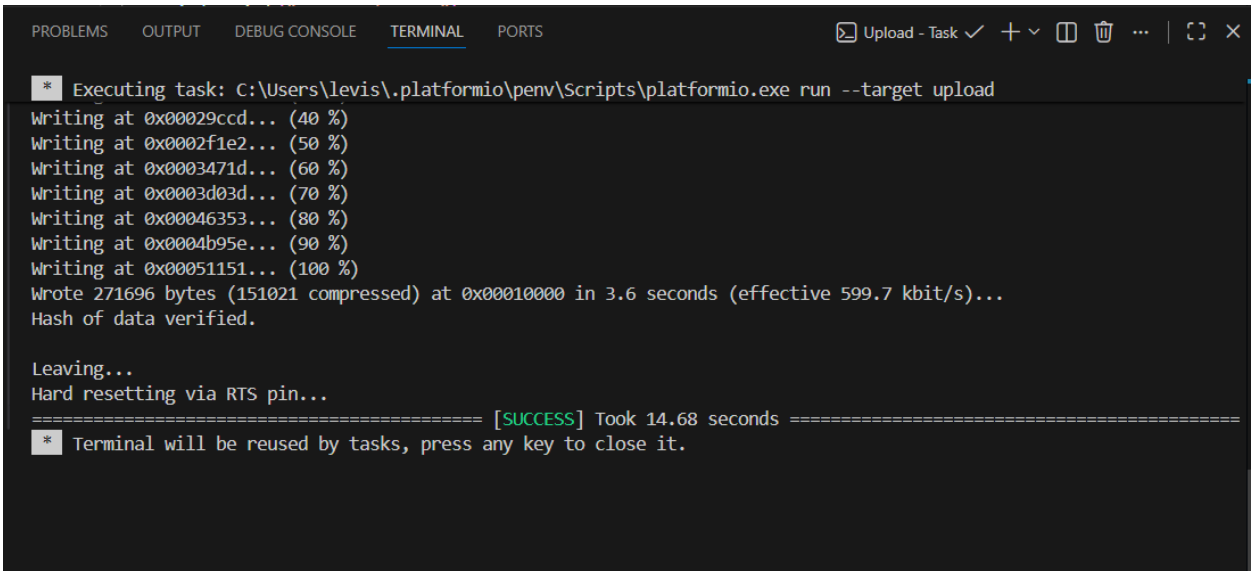
mathworks/ThingSpeak @ ^2.1.1
adafruit/DHT sensor library @ ^1.4.7
Terminal will be reused by tasks, press any key to close it.
Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe run
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
[SUCCESS] Took 15.89 seconds
Terminal will be reused by tasks, press any key to close it.


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Upload - Task
Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe run --target upload
Writing at 0x00029ccd... (40 %)
Writing at 0x0002f1e2... (50 %)
Writing at 0x0003471d... (60 %)
Writing at 0x0003d03d... (70 %)
Writing at 0x00046353... (80 %)
Writing at 0x0004b95e... (90 %)
Writing at 0x00051151... (100 %)
Wrote 271696 bytes (151021 compressed) at 0x00010000 in 3.6 seconds (effective 599.7 kbit/s)...
Hash of data verified.
Leaving...
Hard resetting via RTS pin...
[SUCCESS] Took 14.68 seconds
Terminal will be reused by tasks, press any key to close it.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

* Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe device monitor
--- Available filters and text transformations: debug, default, direct, esp32_exception_decoder, hexl
ify, log2file, nocontrol, printable, send_on_enter, time
--- More details at https://bit.ly/pio-monitor-filters
--- Quit: Ctrl+C | Menu: Ctrl+T | Help: Ctrl+T followed by Ctrl+H
PI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1184
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4
Sensor de Temperatura y Humedad iniciado
Error leyendo el sensor
Error leyendo el sensor
Error leyendo el sensor
Error leyendo el sensor
Error leyendo el sensor
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TE

* Executing task: C:\Users\levis\.p
----------------------
Temperatura: 0.50 °C
Humedad: 20.60 %
----------------------
Temperatura: -1.00 °C
Humedad: 20.10 %
----------------------
Temperatura: -1.00 °C
Humedad: 19.70 %
----------------------
Temperatura: -0.90 °C
Humedad: 19.60 %
----------------------
Temperatura: -0.90 °C
Humedad: 19.90 %
----------------------
```

```cpp
#include <Arduino.h>
#include <DHT.h>

#define DHTPIN 21        // Pin donde conectas DATA
#define DHTTYPE DHT22   // Cambia a DHT22 si usas ese sensor

DHT dht(DHTPIN, DHTTYPE);

void setup() {
    Serial.begin(115200);
    dht.begin();

    Serial.println("Sensor de Temperatura y Humedad iniciado");
}

void loop() {

    float temperatura = dht.readTemperature();
```

```
Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe device monitor
----------------------
Temperatura: 22.70 °C
Humedad: 72.50 %
----------------------
Temperatura: 22.80 °C
Humedad: 72.40 %
----------------------
Temperatura: 22.70 °C
Humedad: 72.30 %
----------------------
Temperatura: 22.80 °C
Humedad: 72.30 %
----------------------
Temperatura: 22.80 °C
Humedad: 72.30 %
----------------------
```

```cpp
    Serial.print("Temperatura: ");
    Serial.println(temperatura);

    Serial.print("Humedad: ");
    Serial.println(humedad);

    // Enviar datos
    ThingSpeak.setField(1, temperatura);
    ThingSpeak.setField(2, humedad);

    int respuesta = ThingSpeak.writeFields(channelID, writeAPIKey);
```

```
Executing task: C:\Users\levis\.platformio\penv\Scripts\platformio.exe device monitor
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4

Brownout detector was triggered

ets Jul 29 2019 12:21:46

rst:0xc (SW_CPU_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1184
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4

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mode:DIO, clock div:2
load:0x3fff0030,len:1184
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4
```

```
Brownout detector was triggered

ets Jul 29 2019 12:21:46

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rst:0xc (SW_CPU_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
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clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1184
load:0x40078000,len:13232
load:0x40080400,len:3028
entry 0x400805e4
```

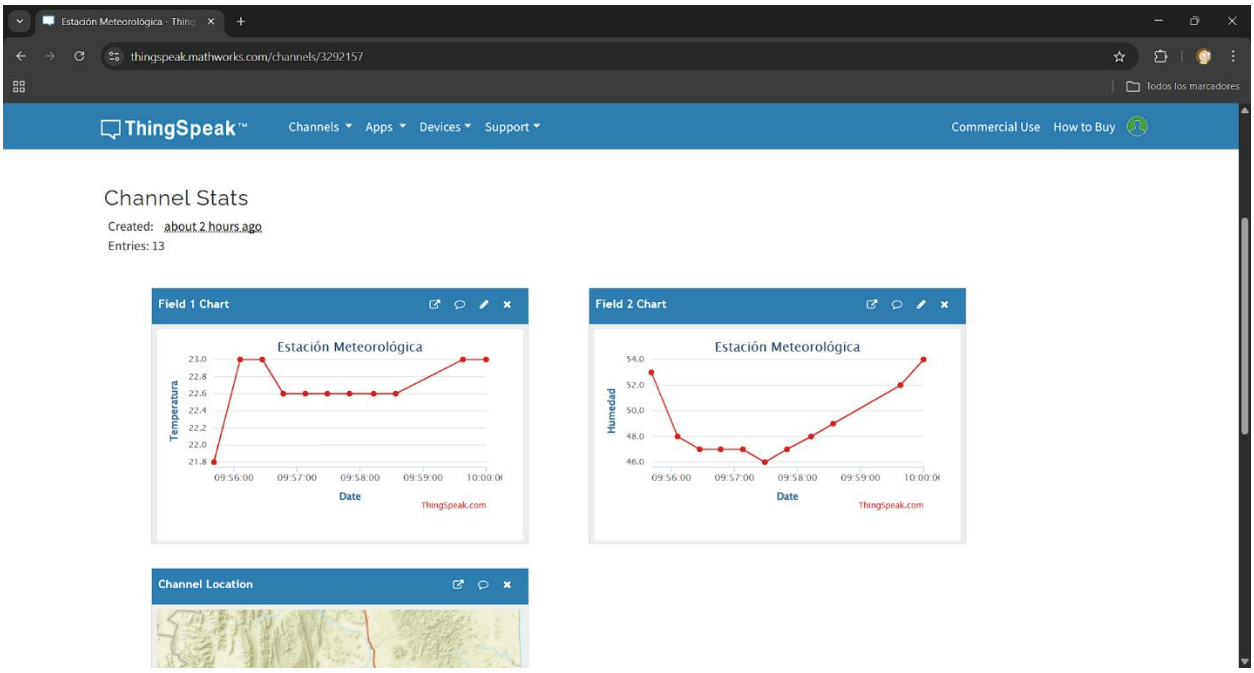
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Entries: 13
Field 1 Chart
Estación Meteorológica
Temperatura
23.0
22.8
22.6
22.4
22.2
22.0
21.8
09:56:00
09:57:00
09:58:00
09:59:00
Date
ThingSpeak.com
Field 2 Chart
Estación Meteorológica
Humedad
54.0
52.0
50.0
48.0
46.0
09:56:00
09:57:00
09:58:00
09:59:00
Date
ThingSpeak.com
Channel Location